Needs Review Agent — Build Scope

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What we're building

An AI agent that does the 1st-layer Needs Review work the service team does by hand today, for Inland Marine applications that SEON flagged or that rent $100K or more of equipment. It answers one question only: **is the client real?** It does not judge character, and it does not hold up an account to check exact limits once one rental house gives a strong verification.

For each case the agent:

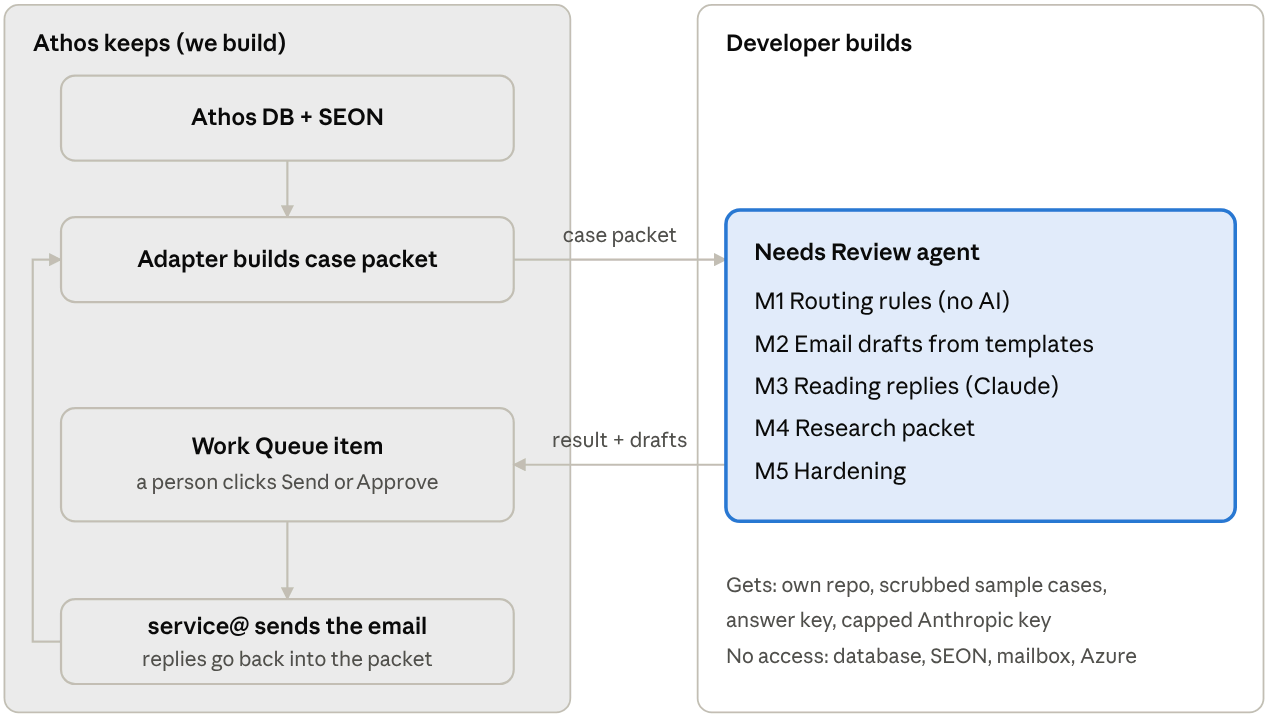
1. Reads the SEON result and the application, and picks the path from the workflow doc.
2. Drafts the emails that path calls for: rental-house vetting (Exhibit 1), applicant references (Exhibit 2), US-residency confirmation.
3. Reads the replies as they arrive and decides whether the client is verified.
4. When verification fails, assembles a research packet (web search, domain age, social links) so a person can choose between decline and Stripe Selfie ID.

It is a first project for a new developer, scoped so they never touch Athos systems. Where it ends up (Advanced Admin or beside the Athos Underwriting AI tool) can be decided later.

Why the developer needs no system access

The agent is a service that keeps no data between calls. Each call sends it one JSON case packet with everything known so far, and it returns what to do next. A case waiting three days on rental houses is just a packet with three emails sent and none received, so every scenario can be tested from a file.

The developer builds only the agent; Athos keeps every system



who builds what · one packet in, one result out

	Athos	Developer
Athos database, staging or production	Yes	No
SEON account and API key	Yes	No: scrubbed SEON responses in the sample cases
service@ mailbox, Graph, SendGrid	Yes	No: returns drafts, never sends
Azure, Key Vault, existing repos	Yes	No: own repo only
Anthropic API key	Issues one with a monthly cap	Uses it
Case history across days	Work Queue item and stored emails	None; each call stands alone
Final approve or decline	A person, in the Work Queue	Never; recommends only

The contract

The agent takes one JSON case packet and returns one JSON result. This is version 0: the developer refines it in M1, and we sign off before M2.

Case packet (in):

Field	What it holds
caseId, product	Case id; IM annual or IM short-term
applicant	Name, business name, email, phone, address, websites and social links they gave
rentedEquipmentLimit	Requested rented equipment limit, USD
underwritingGates	Gates already hit: OFAC, Mexico, RCV over $150K, claims, trailers, TIB

Field	What it holds
seon	State (approve / review / decline), fraud score, fraud reasons, and the full SEON response when we have it
rentalHouses	Each rental house: name, individual or business, contacts, stated equipment value
correspondence	Every email sent and received so far, oldest first, tagged by party and template
now	The current time, so tests control how long the case has been waiting

Result (out):

Field	What it holds
stage	Triage, awaiting rental houses, awaiting applicant, verified, escalated, done
recommendation	Continue, route to underwriting, send emails, wait, approve, escalate to service@, recommend decline
reasons	Each reason with the packet field or source URL it rests on
outbound	Drafted emails: recipients, template, subject, body
extracted	What each reply said: worked with client before, history, replacement cost value, GL required, signed contract, references given, and a strength (strong, weak, none, negative)
verifiedEquipmentTotal	Equipment value tallied from replies; does not block approval
researchPacket	Only on escalation: findings, each with a source URL
followUpAt	When to look again if nothing arrives

Milestones

Each milestone is accepted when the developer's test harness matches underwriting's labelled answers on the sample cases. This is the same fixture pattern the Athos Underwriting AI tool uses.

#	Milestone	What it does	Accepted when	Size
M1	Routing rules (no AI)	The workflow's routing as plain, tested code: $100K threshold, SEON approve / review / decline, gate pass-through, foreign-country branch, and the email, phone and IP-location checks. Thresholds live in one config file.	Every triage case routes correctly; final contract delivered as a zod schema	Small
M2	Email drafts	Exhibit 1 per rental-house contact, Exhibit 2, and the US-residency email. Wording is verbatim from underwriting; the agent fills names and picks variants. No AI writes customer-facing text.	Every variant matches its saved snapshot; no draft on a path that shouldn't send	Small
M3	Reading replies (Claude)	Turns rental-house and applicant replies into structured answers, then decides: one strong verification means approve; rental houses can't help means Exhibit 2; questionable references means escalate. Handles forwards, signatures, partial answers, out-of-office.	95% or better agreement with underwriting, and zero cases where the agent says approve and underwriting doesn't	Medium

#	Milestone	What it does	Accepted when	Size
M4	Research packet	For escalations: web search on name, business and city; domain creation date from RDAP (the data behind the ICANN lookup); IMDb search; the applicant's own social links with what to check. No scraping LinkedIn, Instagram or Facebook.	A packet for every escalation case; every finding has a source URL	Medium
M5	Hardening	Replies that try to instruct the agent are scored negative, cost and timing logged per call, Dockerfile, an HTTP endpoint and a command line that runs one case file.	Injection case passes; runs in Docker	Small

Out of scope for the developer: sending email, reading the mailbox, any database, any screen beyond the test harness, and deciding underwriting gates (OFAC, Mexico and the rest pass through untouched).

What the developer gets on day one

1. This doc and the underwriting workflow doc.
2. A new empty repo with push access. Nothing else in Bitbucket.
3. About 25 sample cases ("fixtures"): JSON case packets copied from real Needs Review cases, with every name, email, phone, address, IP and SEON id replaced by fake values. The shape of the SEON data stays real. The mix:
    - Clean $100K+ case; SEON review, domestic; SEON review, foreign; SEON decline with confirmed fraud
    - Under $100K flagged, looks real; under $100K flagged, looks fake; underwriting gates present
    - Rental-house replies: strong, weak, "never heard of them", out-of-office, forwarded chain

    - Applicant reference replies; no reply after several days; one reply that tries to instruct the agent
4. The right answer for each case, filled in by underwriting. These are the answer key the build is graded against.
5. An Anthropic API key with a monthly spending cap.
6. Suggested stack: TypeScript, Express, zod, the Anthropic SDK and jest, the same as the Athos Underwriting AI tool, so the agent can later sit inside that tool or be called from Advanced Admin without a rewrite.

Before work starts, the engagement agreement should make the code work-for-hire owned by Athos and cover the sample data, even though it is scrubbed.

Athos-side work (ours, not the developer's)

The sample cases are the long pole, so they start first. The SEON data gap needs a .NET handoff before the agent can run on live cases.

- [ ] **Build the sample cases and get them labelled by underwriting.**
- [ ] **Keep the full SEON response.** Today we save only four values on `Customer`: verified flag, score, blackbox score and fraud reasons. The email account counts, phone account counts and IP location this workflow uses live only in the SEON report email and the SEON dashboard. Either store the raw response (a database change, so a .NET handoff) or look it up from SEON at review time, if their API allows it (not yet checked).
- [ ] **Adapter:** build the case packet from the database: customer, the rental houses saved by the 2nd-layer review form, the IM gates, the stored SEON response and stored emails.
- [ ] **Work Queue lane:** show the agent's recommendation, reasons and drafted emails on the Needs Review item. A person clicks Send, Approve or Escalate. The item is still marked done by database state, not by the agent.
- [ ] **Send and capture:** send approved drafts from service@ and attach replies to the case. The Underwriting AI tool already watches a mailbox through Microsoft Graph; the same pattern can capture rental-house replies.
- [ ] **Shadow pilot:** run the agent on live cases for two to four weeks while people keep working them by hand, and compare before anyone acts on its recommendation.

Open questions for underwriting

The workflow doc reads clearly to a person, but these ten points need a firm answer before the routing rules in M1 are final.

- [ ] **Strong verification:** what counts? Proposed: the rental house has rented to this client before, or a person they know referred the client and they're comfortable. Is "we checked references" alone weak?
- [ ] **US-residency email:** the doc says to send one but has no template. We need its wording (an Exhibit 3).
- [ ] **Exact thresholds:** does "over 5" email accounts mean 6 or more? "Over 2" phone accounts, 3 or more? How far apart is "far away" for IP location vs. address: a number of miles, a different state, or a different country?
- [ ] **Under $100K, flagged, looks fake:** is the next step Exhibit 1 to the rental houses, same as the $100K+ path?
- [ ] **Exhibit 1 wording:** when is it "first time working with this client" vs. "requesting higher limits"? Proposed: higher limits when the client has a prior Athos policy.
- [ ] **Timing:** how long do we wait on rental houses before sending Exhibit 2? Is there a reminder in between?
- [ ] **Client says they are not a US resident or entity:** decline?
- [ ] **Products:** IM short-term and annual only, or also STP and EGL applications that list rental houses?
- [ ] **Partial verification:** after one strong verification, is the policy bound at the requested limit or at the verified total?
- [ ] **Who acts on the recommendation:** anyone on staff in the Work Queue, or underwriters only?

Guardrails for v1

- The agent never sends an email, never approves and never declines. It recommends; a person acts.
- Text sent to customers and rental houses comes only from approved templates.
- Every reason cites a packet field or a source URL.
- Replies are untrusted. Instructions inside them are ignored and flagged.
- No real customer data leaves Athos systems during the build. Production data reaches Claude only after go-live, through our adapter and our Anthropic account.